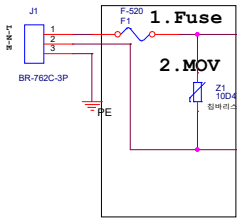
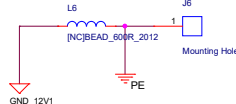
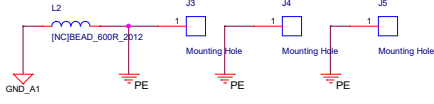
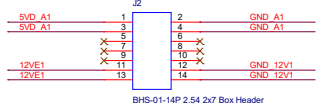
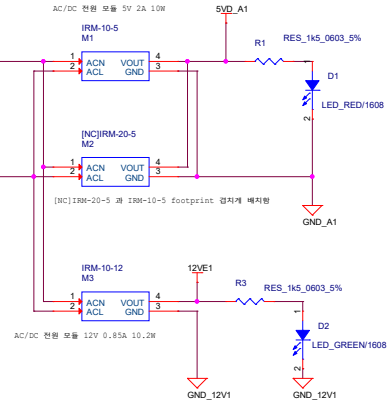
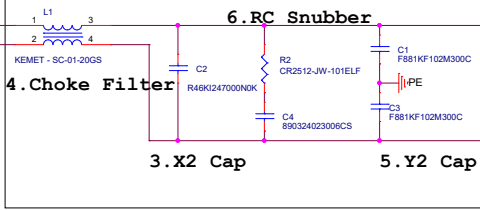


AC DC Converter

필수 구성요소



선택 가능 요소



** 전원 노이즈 보호 회로 순서

퓨즈(Fuse) → 과전류·단락 발생 시 회로를 차단하여 화재와 2차 손상을 방지하기 위해 사용

MOV (Metal Oxide Varistor) → 낙뢰·스태킹으로 발생하는 순간 서지를 흡수하여 입력단 부품을 보호하기 위해 사용

Choke Filter (Common Mode Choke) → 전원선상 통해 유입·양분되는 공통모드 고주파 노이즈를 억제하기 위해 사용

X2 Capacitor (LineFilter) → LTV 사이의 차동모드 노이즈를 감쇠하고 전원 입력의 고주파 안정성을 높이기 위해 사용

RC Snubber → 스위칭이나 인덕터 차단 시 발생하는 급격한 dv/dt와 스파이크를 완화해 EMI를 줄이기 위해 사용

Y2 Capacitor (LineToEarth) → LTV에서 세시(임지)로 흐르는 공통모드 노이즈를 안전하게 우회시켜 EMI를 감소시키기 위해 사용

** 비용 절감 시 제거 우선순위

RC Snubber

→ 특정 부하(인덕터·모터)가 없으면 효과 대비 비용·공간 부담이 큼

Y2 Capacitor

→ EMI 성능을 떨어뜨리지만, 비접지(Class II) 구조나 저노이즈 환경에서는 생략 가능

Choke Filter (Common Mode Choke)

→ 단가가 비교적 높아 초기 EMC 테스트 전에는 제거 대상 1순위

X2 Capacitor

→ 차동모드 노이즈 억제에 중요하지만, 최소 구성에서는 생략 가능

MOV

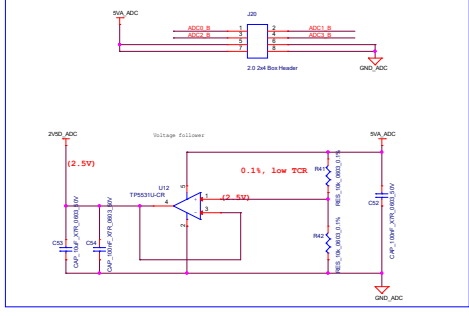
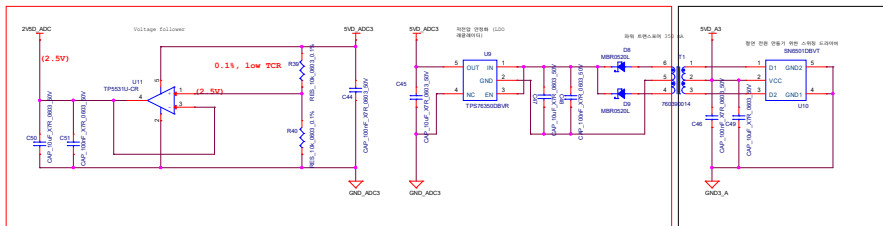
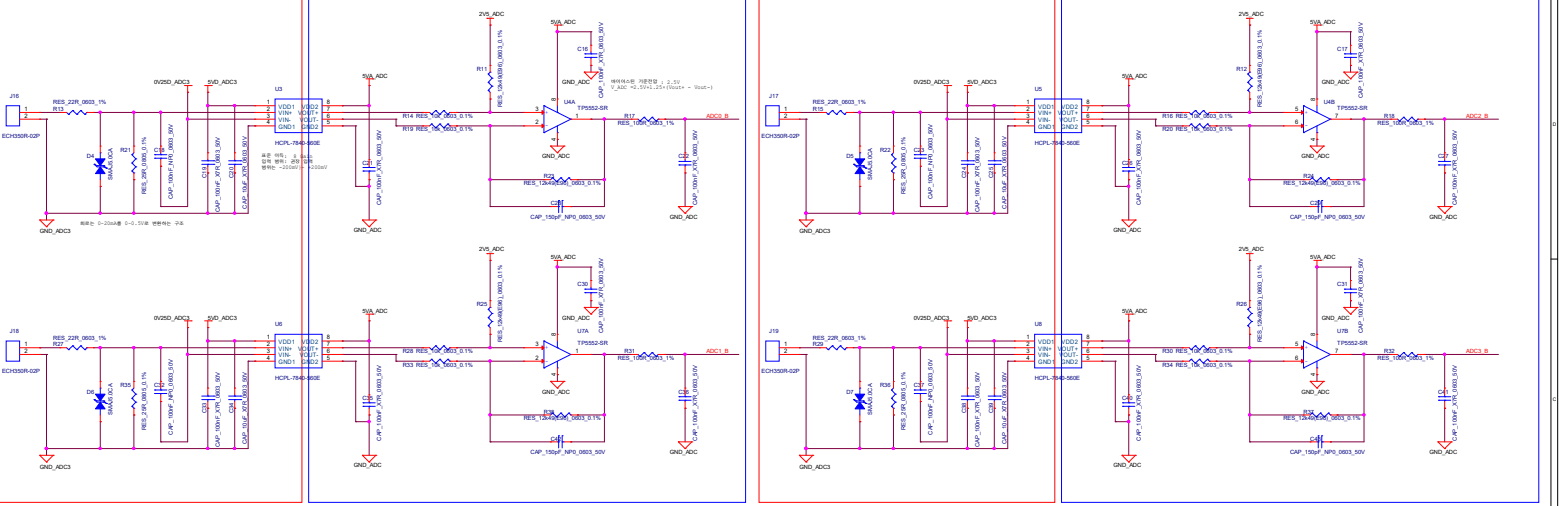
→ 서지 보호 핵심 부품으로, 현장 환경이 불확실하면 유지 권장

퓨즈(Fuse)

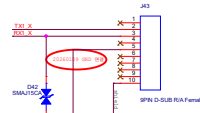
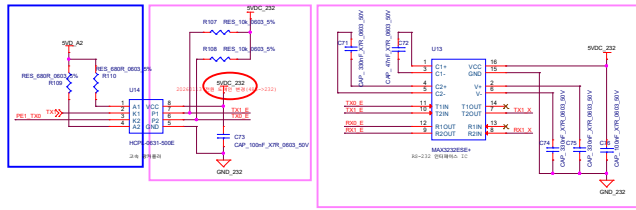
→ 안전 규정·화재 방지 핵심 부품 → 비용과 무관하게 최우선 유지

PE : Protect Eearth
FG : Frame GND
SHGND : Shield GND
DGND : Digital GND
SGND : Signal GND
AGND : Analog GND

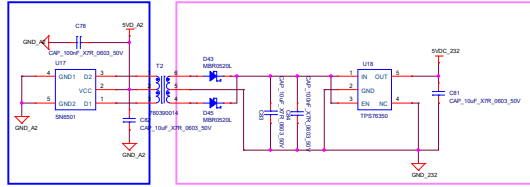
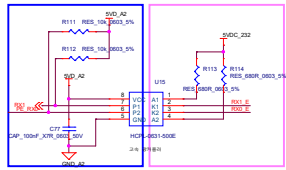
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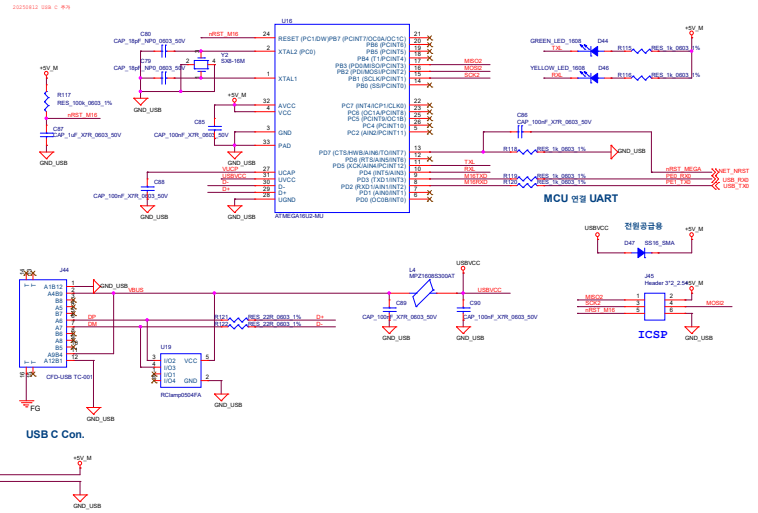
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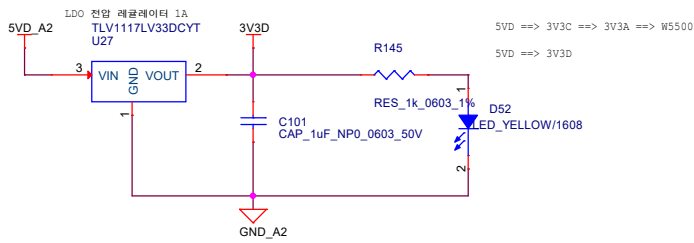
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 FG : Frame GND
 SHGND : Shield GND
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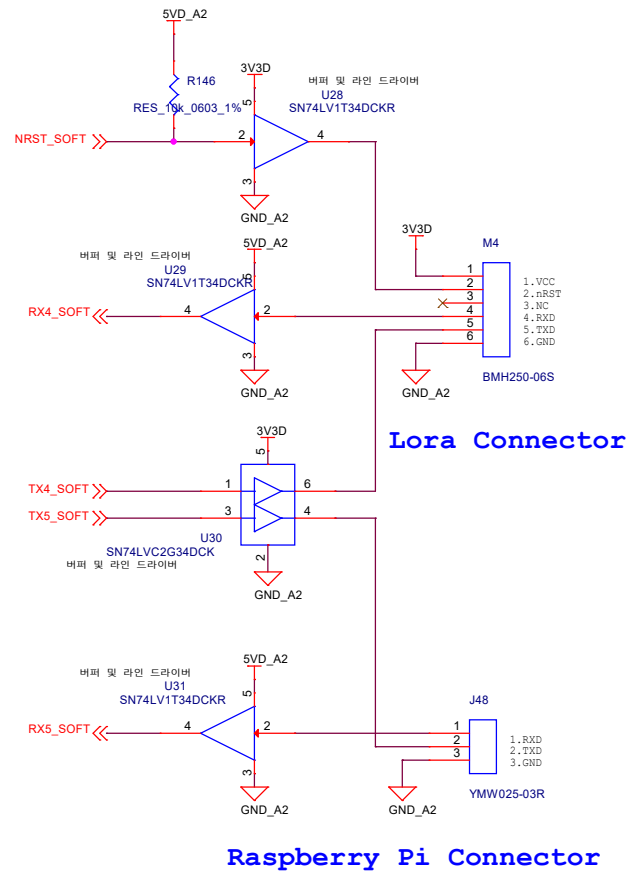
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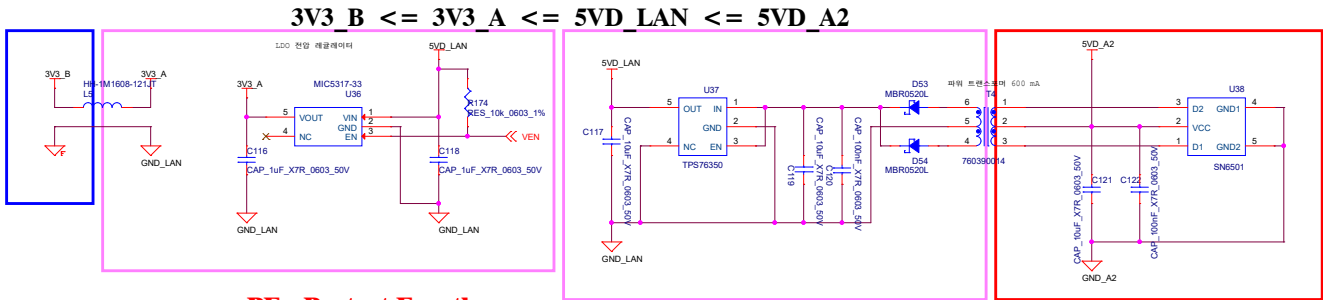
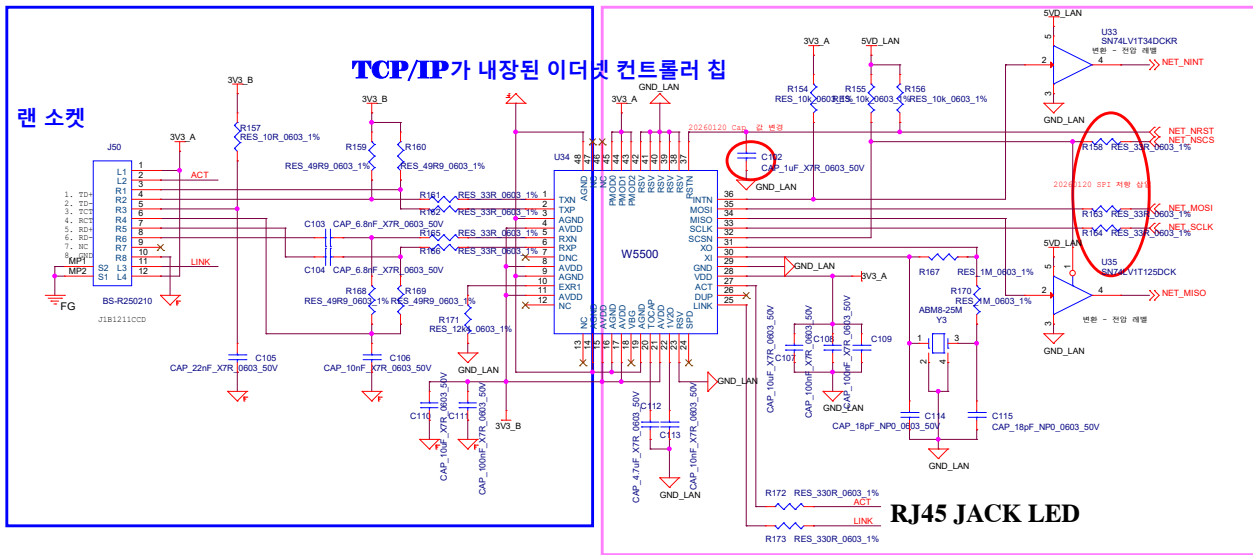
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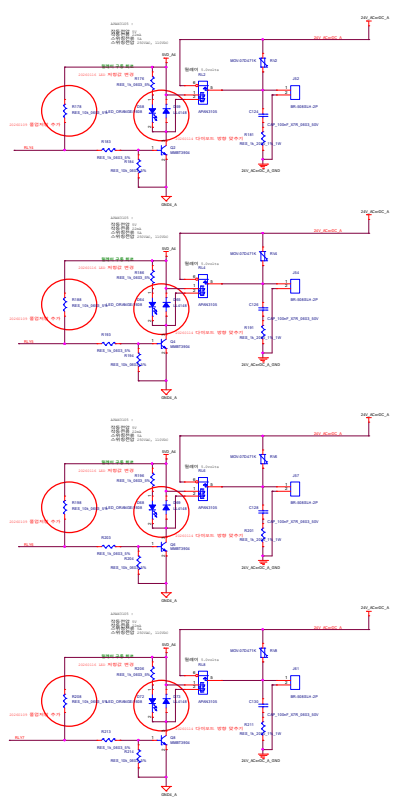
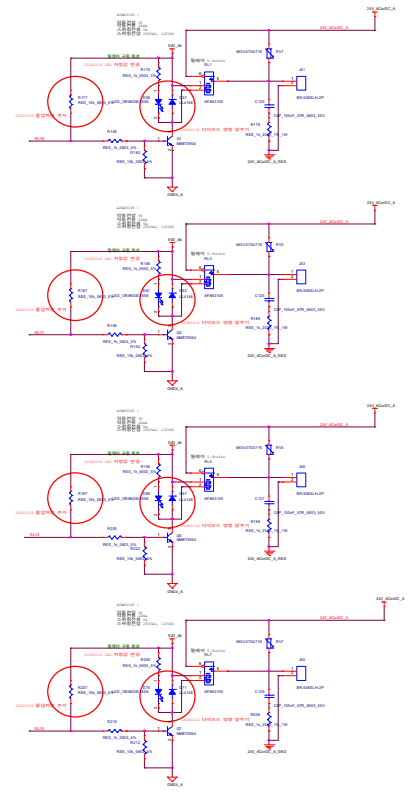
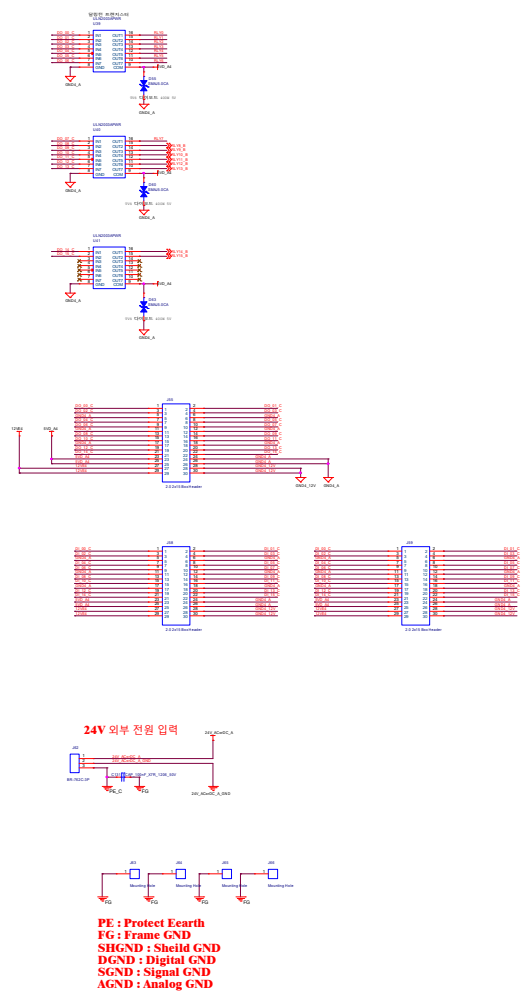


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Sheet 7 of 11		



PE : Protect Eearth
FG : Frame GND
SHGND : Sheild GND
DGND : Digital GND
SGND : Signal GND
AGND : Analog GND

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Sheet			8 of 11
Rev			03



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전원선
접지선

